

YONATAN BELINKOV, PH.D.

The Henry and Marilyn Taub Faculty of Computer Science
Technion – Israel Institute of Technology

1. PERSONAL DETAILS

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2. ACADEMIC DEGREES

- 2018 **Ph.D. in Electrical Engineering and Computer Science**, MIT, Cambridge, MA
Thesis: On Internal Language Representations in Deep Learning: An Analysis of Machine Translation and Speech Recognition
Advisor: James Glass, Senior Research Scientist, Computer Science and Artificial Intelligence Laboratory (CSAIL), and Faculty Member, Harvard-MIT Health Sciences & Technology
- 2014 **M.A. in Arabic and Islamic Studies** (*summa cum laude*), Tel Aviv University, Israel
- 2009 **B.Sc. in Mathematics** (*magna cum laude*) and **Arabic and Islamic Studies** (*summa cum laude*), Tel Aviv University, Israel

3. ACADEMIC APPOINTMENTS

- 2025–26 **Visiting Scholar**, Kempner Institute, Harvard University
- 2025– **Associate Professor**, Faculty of Computer Science, Technion, Haifa, Israel
- 2020–25 **Senior Lecturer** (Assistant Professor), Faculty of Computer Science, Technion, Haifa, Israel
- 2018–20 **Postdoctoral Fellow in Computer Science**, SEAS, Harvard University, Cambridge, MA
Faculty Host: Stuart Shieber, Professor of Computer Science
- 2018–20 **Postdoctoral Associate in Computer Science**, CSAIL, MIT, Cambridge, MA
Faculty Host: James Glass, Senior Research Scientist, CSAIL, and Faculty Member, Harvard-MIT Health Sciences & Technology

4. PROFESSIONAL EXPERIENCE

- 2010–12 Software Engineer, Intuview Ltd., Israel

5. RESEARCH INTERESTS

Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Interpretability, Robustness, Emergent Communication, Biological Language Models.

6. TEACHING EXPERIENCE

Lecturer, Technion, Haifa, Israel

- CS236004: Introduction to Transformers (Fall 2023, Spring 2025)
- CS236299: Introduction to Natural Language Processing (Fall 2020, Spring 2022, Spring 2023, Fall 2024)
- CS236756: Introduction to Machine Learning (Fall 2021, Fall 2022, Fall 2023, Fall 2024)
- CS236817: Seminar in Natural Language Processing (Spring 2021, Fall 2021, Fall 2022)

Co-Instructor, MIT, Cambridge, MA (2020)

- Structure and Interpretation of Deep Networks

Co-Instructor, Harvard University, Cambridge, MA (2019)

- Curricular Design for Computer Science: Computational Linguistics and Natural-language Processing

Lecturer, Tel Aviv University, Israel

- Fundamentals of Grammar (2009–2011)
- Arabic II (2009–2011)
- Grammar I (2010)

Teaching Assistant, MIT, Cambridge, MA (2015)

- Introduction to Computer Science and Programming in Python
- Introduction to Computational Thinking and Data Science

Guest Lecturer

- Natural Language Processing, Princeton, Princeton, NJ (2021)
- Advanced Natural Language Processing, MIT, Cambridge, MA (2020)
- Language, Structure, and Cognition, Harvard, Cambridge, MA (2019)
- Automatic Speech Recognition, MIT, Cambridge, MA (2019)
- Machine Translation and Sequence-to-sequence Models, CMU, Pittsburgh, PA (2018)
- NLP and the Humanities, Hebrew University, Jerusalem, Israel (2015)

Pedagogical Training, MIT, Cambridge, MA (2015)

Kaufman Teaching Certificate Program, Teaching and Learning Laboratory

7. TECHNION ACTIVITIES

8. DEPARTMENTAL ACTIVITIES

- 2023–25 Faculty Website, Computer Science, Technion
- 2022–23 Counselor for students with an unsatisfactory academic status, Computer Science, Technion
- 2021–25 Faculty Search Committee, Computer Science, Technion
- 2015–18 Graduate Admissions Committee Member, EECS, MIT

9. PUBLIC PROFESSIONAL ACTIVITIES

Workshop Organizer

BlackboxNLP (at *ACL* 2019, *EMNLP* 2020, *ACL* 2021, *EMNLP* 2022, *EMNLP* 2023, *EMNLP* 2024, and *EMNLP* 2025), *Robustness Task* (at *WMT* 2019 and *WMT* 2020), *RobustML* (at *ICLR* 2021), *NLP and DH: Hebrew and other languages* (2024), *Invented, Constructed, and Emergent Languages in Multi-agent AI Systems* (at MIT and Harvard, 2026)

Senior Area Chair

Interpretability and Analysis of Models for NLP track at *EMNLP* (2023), *Interpretability and Analysis of Models for NLP track* at *NAACL* (2021, 2025), *Interpretability and Analysis of Models for NLP track* at *ACL* (2022, 2025), *Ethical and Sustainable NLP track* at *EACL* (2023)

Area Chair

Interpretability and Analysis of Models for NLP track at *ACL* (2020, 2021), *Interpretability and Analysis of Models for NLP track* at *EMNLP* (2020, 2021), *CoNLL* (2020), *ACL ARR* (May 2025), *NeurIPS* (2021, 2022), *COLM* (2024, 2025)

Reviewer

- **Journals:** *Computational Linguistics* (2021, 2022), *TACL* (2020–2022), *IEEE TASL* (2014, 2016, 2018), *Computer Speech and Language* (2017), *ACM Surveys* (2022)
- **Conferences:** *ACL Rolling Review* (2021, 2025, 2026), *ACL* (2018, 2019, 2023), *EMNLP* (2015, 2017, 2018 [best reviewer], 2019, 2022, 2025 [best paper committee]), *NAACL* (2018, 2019), *NeurIPS* (2019, 2020, 2024), *ICLR* (2019 [outstanding reviewer], 2020, 2021 [outstanding reviewer], 2022, 2025), *EACL* (2021), *Coling* (2018 [outstanding reviewer]), *CoNLL* (2016–2018, 2021), *IJCAI* (2019)
- **Workshops:** Various NLP workshops
- **Grant proposals:** Israeli Science Foundation (2021), Hasler Foundation (2021), Swiss National Science Foundation (2022), Czech Science Foundation (2022), Israeli Ministry of Science and Technology (2022, 2026), New Zealand Marsden Fund (2023), ERC Starting (2025)

Tutorial Instructor

Tutorial on *Interpretability and Analysis in Neural NLP* at *ACL* (2020) (video)

ArXiv moderator

Moderator of the cs.CL subject on ArXiv (2020–2025)

10. MEMBERSHIP IN PROFESSIONAL SOCIETIES

- The Association for Computational Linguistics (since 2015)
- ELLIS Scholar (since 2021)
- The Israeli Association for Artificial Intelligence (2025)

11. FELLOWSHIPS, AWARDS, AND HONORS

Fellowships

- 2020–23 Azrieli Early Career Faculty Fellowship
- 2020–23 Viterbi Fellowship, Center for Computer Engineering, Technion
- 2018–20 Mind, Brain, and Behavior Postdoctoral Fellowship, Harvard University
- 2018 Moore-Sloan Data Science Fellow, NYU (*declined*)

Awards and Honors

2026	The Cooper Award for Excellence in Research, Technion
2025	Outstanding paper award, Conference on Empirical Methods in Natural Language Processing
2025	Krill Prize for Excellence in Scientific Research, Wolf Foundation
2024	Best Paper Award, Conference on Empirical Methods in Natural Language Processing
2024	Henry Taub Prize for Academic Excellence, Technion
2023	Excellence in Teaching, Introduction to Machine Learning, Technion
2021	AAAI New Faculty Highlights Program, AAAI
2019	ICLR Travel Award, New Orleans, LA
2017	NeurIPS Travel Award, Long Beach, CA
2016	Coling Student Support Program, Osaka, Japan
2013	Elie Shaio Memorial Award, MIT
2012	Konard Adenauer Master’s Thesis Scholarship, Tel Aviv University
2010–11	Puzis Academic Achievements Award, Faculty of Humanities, Tel Aviv University
2010	The Rina Drori Excellence Scholarship, Faculty of Humanities, Tel Aviv University
2010	Excellence scholarship, Department of Arabic and Islamic Studies, Tel Aviv University
2009	Excellence Scholarship, Wolf Foundation
2009	Excellence Award, School of Mathematical Sciences, Tel Aviv University

12. GRADUATE STUDENTS

Completed PhD theses

- Boaz Carmeli, PhD student, Technion, *On Artificial Agents that Learn to Communicate* (2022–2026) (Co-advisor with Ron Meir)
- Edo Dotan, PhD student, Technion, *Deep Learning for Biology* (2023–2026) (Co-advisor with Tal Pupko)
- Hadas Orgad, PhD student, Technion, *On the Robustness of AI Models: Explanations, Evaluations, and Improvements* (2022–2025; Apple AI/ML PhD Fellowship; Runner Up for the IAAI Best Dissertation Award; Blavatnik Computer Science PhD Award)

Completed MSc theses

- Ido Levy, MSc student, Technion, *Unsupervised Translation of Emergent Communication* (2023–2025)
- Yanay Soker, MSc student, Technion, *Predicting Success of Model Editing Through Intrinsic Features* (2023–2024)
- Rotem Ben-Zion, MSc student, Technion, *Semantics and Spatiality of Emergent Communication*, (2023–2024)
- Zachary Bamberger, MSc student, Technion, *DEPTH: Discourse Education through Pre-Training Hierarchically* (2022–2024)
- Shadi Iskander, MSc student, Technion, *Mitigating Social Bias in Language Models: Fairness Strategies in Labeled and Unlabeled Demographics Settings* (2022–2024) (Co-advisor with Kira Radinsky)
- Shachar Katz, MSc student, Technion *Visualizing and Interpreting the Semantic Information Flow of Transformers* (2022–2024)
- Reda Igbaria, MSc student, Technion, *Debiasing Natural Language Understanding Models Through Biased Internal Components* (2021–2023)
- Omer Antverg, MSc student, Technion, *Analyzing Individual Neurons in Contextual Word Representations from Neural Language Models* (2021–2022)

- Michael Mendelson, MSc student, Technion, *How Debiasing Affects Internal Representations in Natural Language Understanding Models* (2020–2021)
- Yana Dranker, MSc student, Technion, *Invariant Risk Minimization for Natural Language Inference* (2020–2022)
- Dimion Asael, MSc student, Technion, *A Generative Approach for Mitigating Structural Biases in Natural Language Inference* (2020–2021)
- Michal Kessler, MSc student, Hebrew University, *Machine Learning for Judeo-Arabic* (2019–2021) (Co-advisor with Omri Abend)
- Rami Manna, MEng student, MIT, *Low Resource Speech-to-text Translation from Arabic to English* (2019–2021) (Co-advisor with James Glass)

PhD theses in progress

- Tal Haklay, PhD student, Technion, *Interpretability of LLMs in Real World Settings* (2022–; VATAT Fellowship for Outstanding Women Doctoral Students in Hi-Tech)
- Tomer Ashuach, PhD student, Technion, *Uncovering Mechanisms of Language Models’ Capabilities* (2023–)
- Yaniv Nikankin, PhD student, Technion, *Scaling Up Model Interpretability* (2024–; VATAT PhD Fellowship in AI and Data Science)
- Michael Toker, PhD student, Technion, *Interpretable LLMs: Robustness & Alignment* (2022–; Azrieli PhD Fellowship)
- Dana Arad, PhD student, Technion, *Editing Text-to-Image Models* (2023–; Ariane de Rothschild PhD Fellowship)
- Adi Simhi, PhD student, Technion, *Promoting Trust in AI Outputs* (2023–, VATAT PhD Fellowship in AI and Data Science)
- Adir Rahamim, PhD student, Technion, *Interpreting Language Models Across Time* (2023–)
- Itay Itzhak, PhD student, Technion, *Cognitive Biases of Language Models* (2023–) (Co-advisor with Gabi Stanovsky)
- Michael Hanna, PhD student, University of Amsterdam (2022–; Google PhD Fellowship) (Co-advisor with Sandro Pezzelle)

MSc theses in progress

- Gal Kesten, MSc student, Technion, *Interpretability of Protein Language Models* (2025–)

PhD thesis reader / committee member

- Assaf Ben-Kish, PhD student, Tel Aviv University, *Improving Large Language Models* (2025) (PhD committee member)
- Zachary Bamberger, PhD student, Technion, *Generating and Evaluating Persuasive Arguments with Large Language Models* (2025) (PhD committee member)
- Liran Ringel, PhD student, Technion, *Reliable and Efficient Language Models* (2025) (PhD committee member)
- Yara Shamshoum, PhD student, Technion, *Efficiency and Reasoning in AI Models* (2025) (PhD committee member)
- Sahar Admoni, PhD student, Technion, *Leveraging Large Language Models to Interpret Reinforcement Learning Agents* (2025) (PhD committee member)
- Mati Shufan, PhD student, Technion, *Cognitive Traits of the Human Mind as an Intentional System in Interactions with AI Personas* (2025) (PhD committee member)
- Guy Azran, PhD student, Technion, *Decomposing Task and Motion Planning into Learnable Components: Towards Generalizable Robotic Intelligence* (2025) (PhD committee member)
- Zorik Gekhman, PhD student, Technion, *Towards Robust and Trustworthy NLP Models* (2024) (PhD committee member)
- Tomasz Limisiewicz, PhD student, Charles University, *Interpreting and Controlling Linguistic Features in Multilingual Language Models* (2024) (PhD thesis reader)

- Tsachi Blau, PhD student, Technion, *Threat-Model Agnostic Defenses* (2024) (PhD committee member)
- Eyal Cohen, PhD student, Technion, *Smarter Decoders for Automatic Speech Recognition* (2024) (PhD committee member)
- Alon Jacovi, PhD student, Bar Ilan University, *Explaining Artificial Intelligence: Foundations and Practice* (2024) (PhD thesis reader)
- Hanqi Yan, PhD student, University of Warwick, *Enhancing Robustness and Interpretability in Natural Language Processing through Representation Learning* (2024) (PhD thesis reader)
- Adi Haviv, PhD student, Tel Aviv University (2024) (PhD committee member)
- Eilam Shapira, PhD student, Technion (2024) (PhD committee member)
- Aaron Mueller, PhD student, Johns Hopkins University (2023) (PhD committee member)
- Shalev Shaer, PhD student, Technion (2022) (PhD committee member)
- Yoav Levine, PhD student, Hebrew University, *Theoretical Insights on the Application of Deep Neural Networks in the Fields of Many-Body Quantum Physics and Natural Language Processing* (2022) (PhD thesis reader)
- Ido Galil, PhD student, Technion (2022) (PhD committee member)
- Damián Pascual Ortiz, PhD student, ETH Zurich, *Leveraging and Understanding Deep Learning Models from Brain Activity to Language Processing* (2022) (PhD thesis reader)
- James M. Fiocco, PhD student, Carnegie Mellon University, *Functional Components as a Paradigm for Neural Model Design and Explainability* (2022) (PhD committee member)
- Naomi Saphra, PhD student, University of Edinburgh, *Training Dynamics of Neural Language Models* (2021) (PhD thesis reader)

Master’s thesis reader

- Tomer Krichli, Master’s student, Technion, *Adapting a Non-Causal Pre-Trained Speech Recognition Model for Streaming Tasks* (2026)
- Tal Swisa, Master’s student, Technion, *Concept-Based Explicit Modeling of Narratives* (2026)
- Gil Denekamp, Master’s student, Technion, *Learning to learn in quantized-aware training* (2026)
- Zvi Natan Badash, Master’s student, Technion, *Predictive and Uncertainty-Aware Disease Modeling from Unstructured Radiology Reports* (2025)
- Yadin Benyamin, Master’s student, Weizmann Institute, *Emotional Speech Synthesis (ESS) using a large pre-trained Text-to-Speech (TTS) Deep Learning Model* (2025)
- Omer Vered, Master’s student, Technion, *GNN-based Text Representations for Style-focused Personalized LLMs* (2025)
- Oz Huly, Master’s student, Technion, *Predicting RAG Performance for Text Completion* (2025)
- Inbar Nachmani, Master’s student, Technion, *A Proximity Aware Loss Function for Ordinal Classification* (2025)
- Niv Kook, Master’s student, Technion, *Benchmarking Confounder Suggestion Capabilities of Large Language Models in Observational Studies* (2025)
- Daniela Gottesman, Master’s student, Tel Aviv University, *Estimating Knowledge in Large Language Models Without Generating a Single Token* (2024)
- Niv Kook, Master’s student, Technion, *Benchmarking Confounder Suggestion Capabilities of Large Language Models in Observational Studies* (2024)
- Daniel Weisberg Mitelman, Master’s student, Reichman University, *Applying Language Models to Phylogenetic Linguistics and Transliteration* (2024)
- Safaa Shehadi, Master’s student, Haifa University, *Triggers of Code-switching in Arabizi* (2024)
- Yair Gat, Master’s student, Technion, *A Causal Framework for Model Explanations in NLP* (2023)
- Daniel Gilo, Master’s student, Technion, *A General Search-based Framework for Generating Textual Counterfactual Explanations* (2023)
- Dave Makhervaks, Master’s student, Technion, *Clinical Contradiction Detection* (2023)

- Yifan Jiang, Master’s student, University of Washington, *The Weighted Möbius Score: A Unified Framework for Feature Attribution* (2023)
- Shaked Meirom, Master’s student, Technion, *Geometric and Topological approaches for Natural Language Processing* (2023)
- Adi Simhi, Master’s student, Technion, *Interpreting Embedding Spaces by Conceptualization* (2022)
- Ben Finkelshtein, Master’s student, Technion, *Robustness and Rotation Equivariance in Geometric Deep Learning* (2022)
- Mohammed Dabbah, Master’s student, Technion, *Using Fictitious Class Representations to Boost Discriminative Zero-Shot Learners* (2022)
- Itay Itzhak, Master’s student, Tel Aviv University, *Models In a Spelling Bee: Language Models Implicitly Learn the Character Composition of Tokens* (2021)
- Daniel Rosenberg, Master’s student, Technion, *On the Robustness of Visual Question Answering Systems* (2021)
- Gal Sadeh-Kenigsfield, Master’s student, Technion, *Leveraging Auxiliary Text for Deep Recognition of Unseen Visual Relationships* (2021)
- Tomer Wullach, Master’s student, Haifa University, *Generalized Hate Speech Detection on Social Media* (2021)
- Ram Yazdi, Master’s student, Technion, *Perturbation Based Learning for Structured NLP Tasks with Application to Dependency Parsing* (2021)
- Shunit Haviv Hakimi, Master’s student, Technion, *Deep Neural Models for Jazz Improvisations* (2021)
- Elia Turner, Master’s student, Technion, *Charting and Navigating the Space of Solutions for Recurrent Neural Networks* (2021)
- Tom Beer, Master’s student, Technion, *Causal Inference with Mismeasured and Spurious Covariates* (2020)
- Muhammad Majadly, Master’s student, Haifa University, *Dynamic Ensembles in Named Entity Recognition for Historical Arabic Texts* (2020)

Bachelor’s thesis reader

- Victoria Li, BA student, Harvard University, *Conversational AI as Information Mediator* (2026)
- Johnathan Sun, BA student, Harvard University, *The User in the Prompt: A Theory of Context-Sensitive Choice in Language Models* (2026; Winner of the Hoopes Prize)
- Laura Xuan Nguyen, BA student, Harvard University, *Large Language Model Embeddings for Single-Cell Transcriptomics: A Framework for Robust Classification of Motor Neuron Vulnerability* (2026)
- Mirac Suzgun, BA student, Harvard University, *Formal Language Theory as a Framework for Understanding the Limitations of Recurrent Neural Networks* (2020; Winner of the Hoopes Prize)
- Christine Jou, BA student, Harvard University, *Connecting Language Representations in Humans and Machines* (2020)
- Abdul Saleh, BA student, Harvard University, *Towards Social and Interpretable Neural Dialog Systems* (2020)

Other advising experience

- Mentor for seven undergraduate students at MIT (2017–2019)
- Mentor for six undergraduate students at Harvard SEAS (2018–2020)

13. SPONSORED LONG-TERM VISITORS AND POST-DOCTORAL ASSOCIATES

Post-Doctoral Associates

- Anja Reusch, Azrieli Postdoctoral Fellow (2024–)
- Martin Tutek (2024–2025)
- Aaron Mueller, Zuckerman Postdoctoral Scholar (2023–2025) (joint supervision with David Bau)

14. GRANTS

Competitive

- 2026–30 Council for Higher Education Moonshot AI Research Center. *AI for All: Unlocking Scalable and Democratized Foundation Models*. Co-PIs: A. Schuster, D. Soudry, K. Levy, T. Koren, O. Litany, H. Maron, Y. Goldberg, R. Schwartz, G. Chechik, M. Geva, I. Seroussi, and T. Michaeli. Grant amount: 9.88M NIS (approx. \$3.2M total; \$248K per PI).
- 2025–30 Israel Science Foundation Personal Research Grant no. 2942/25. *Scaling Up Interpretability of Language Models*. Grant amount: 1.4M NIS (approx. \$421,000).
- 2025–28 Ministry of Science and Technology Research Grant no. 0008707. *Unlearning Knowledge from Language Models*. Grant amount: 499,983 NIS (approx. \$162,000).
- 2025–28 U.S-Israel Binational Science Foundation Grant no. 2025670. *NSF-BSF: RI: Small: Beyond Binary Variables: Defining, Locating, and Editing Multi-dimensional Features in Large Language Models*. Co-PI: Aaron Mueller. Grant amount: 990,000 NIS (approx. \$298,000).
- 2024–29 European Research Council Starting Grant no. 101165402. *Controlling Large Language Models (Control-LM)*. Grant amount: €1.5M (approx. \$1.66M).
- 2023–27 U.S-Israel Binational Science Foundation Grant no. 2022330. *Emergent Communication in Artificial Agents: Knowledge Disparities and Language Decipherment*. Co-PIs: Shafi Goldwasser and Ron Meir. Grant amount: \$165,200.
- 2021–24 Ministry of Science and Technology Research Grant no. 0002215. *Automatic Detection of Figurative Language in Hebrew across the Eras*. Co-PIs: Benny Kimelfeld and Ophir Münz-Manor. Grant amount: 599,990 NIS (approx. \$189,000).
- 2020–24 Israel Science Foundation Personal Research Grant no. 448/20. *Interpretability and Robustness in Neural Natural Language Processing*. Grant amount: 920,000 NIS (approx. \$270,000).
- 2020–23 Israel Science Foundation New Faculty Equipment Grant no. 449/20. *Interpretability and Robustness in Neural Natural Language Processing*. Grant amount: 647,000 NIS (approx. \$200,000).

Industrial and Other Sources

- 2025–27 Coefficient Giving. *Studying whether LLMs can develop covert communication protocols in referential games*. Grant amount: \$365,378K.
- 2025–27 Coefficient Giving. *Research developing weight pruning methods to remove harmful mechanisms*. Grant amount: \$329,404K.
- 2025–26 Google Gift. *LLM Safety via Weight Pruning*. Grant amount: \$30K.
- 2025–26 EuroHPC AI and Data-Intensive Applications. *Model Editing for Generative Information Retrieval (EHPC-AI-2024A06-058)*. Co-PI: Anja Reusch. Grant amount: 128,000 GPU hours.
- 2024–26 Open Philanthropy Grant. *Support research on interpretability of large language models*. Grant amount: \$184,835.
- 2025–26 IDSAI inter-disciplinary, inter-center research projects. *Text2Protein: Generating Protein Sequences from Rich Textual Descriptions*. Co-PI: Tal Pupko. Grant amount: 100,000 NIS.
- 2024–25 Google Gift. *Full Circuit Finding*. Grant amount: \$30K.
- 2024–25 Pfizer Sponsored Research. *Hallucination Detection in Biomedical Literature Summarization*. Grant amount: \$155,297.
- 2024–25 Google Sponsored Research. *Personas and knowledge in Large Language Models*. Grant amount: \$30K.

- 2024–25 NVIDIA. *Academic Gift Award*. Grant amount: 130,000 NIS (approx. \$35,000).
- 2024–25 IBM Project Agreement. *Local Updates to Language Models*. Grant amount: 120,000 NIS (approx. \$32,000).
- 2022–24 Open Philanthropy Grant. *Initiative for the Interpretable Control of AI*. Co-PI: David Bau. Grant amount: \$1M.
- 2023–24 Google Sponsored Research. *Detecting generated hallucinations within Large Language Models (LLM) and intervening by locating them within the LLM architecture*. Grant amount: \$30K.
- 2020–23 Azrieli Faculty Fellowship Research Grant. *Information Storage in Models of Human Language*. Grant amount: \$209,440.
- 2018–22 International Collaborator on Israel Science Foundation Grant no. 1191/18. *Linguistic Analysis of Algerian Judeo-Arabic Corpora Assisted by Machine Learning*. PI: Ofra Tirosh-Becker, Hebrew University. Grant amount: 520,000 NIS (approx. \$143,000).
- 2019 Harvard Mind, Brain, Behavior Fellow Award. *Language Representations in Humans and Machines* (\$5000).

15. PUBLICATIONS

Theses

- 2018 PhD Thesis: On Internal Language Representations in Deep Learning: An Analysis of Machine Translation and Speech Recognition
Electrical Engineering and Computer Science, MIT, Cambridge, MA
Advisor: James Glass
- 2014 MA Thesis: The Arabic Dialect of Ġisir izZarga: Linguistic description and a preliminary classification, with sample texts
Arabic and Islamic Studies, Tel Aviv University, Israel
Advisor: Nasir Basal

Journal Articles

- [1] Arts, T., **Y. Belinkov**, N. Habash, A. Kilgarrieff, and V. Suchomel. arTenTen: Arabic Corpus and Word Sketches. *Journal of King Saud University – Computer and Information Sciences*. 2014.
- [2] **Belinkov, Y.**, T. Lei, R. Barzilay, and A. Globerson. Exploring Compositional Architectures and Word Vector Representations for Prepositional Phrase Attachment. *Transactions of the Association for Computational Linguistics (TACL)*. 2014.
- [3] Romeo, S., G. Da San Martino, **Y. Belinkov**, A. Barrón-Cedeño, M. Eldesouki, K. Darwish, H. Mubarak, J. Glass, and A. Moschitti. Language processing and learning models for community question answering in Arabic. *Information Processing & Management (IPM)*. 2017.
- [4] Adi, Y., E. Kermany, **Y. Belinkov**, O. Lavi, and Y. Goldberg. Analysis of sentence embedding models using prediction tasks in natural language processing. *IBM Journal of Research and Development*. 2017.
- [5] **Belinkov, Y.** and J. Glass. Analysis Methods in Neural Language Processing: A Survey. *Transactions of the Association for Computational Linguistics (TACL)*. 2019.
- [6] **Belinkov, Y.***, A. Magidow*, A. Barrón-Cedeño, A. Shmidman, and M. Romanov. Studying the History of the Arabic Language: Language Technology and a Large-Scale. *Language Resources and Evaluation*. 2019.
- [7] **Belinkov, Y.***, N. Durrani*, F. Dalvi, H. Sajjad, and J. Glass. On the Linguistic Representational Power of Neural Machine Translation Models. *Computational Linguistics*. 2020.
- [8] **Belinkov, Y.**. Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics*. 2022.
- [9] Tirosh-Becker, O.*, M. Kessler*, O. Becker, and **Y. Belinkov**. Part-of-Speech and Morphological Tagging of Algerian Judeo-Arabic. *Northern European Journal of Language Technology*. 2022.
- [10] Itzhak, I., G. Stanovsky, N. Rosenfeld, and **Y. Belinkov**. Instructed to Bias: Instruction-Tuned Language Models Exhibit Emergent Cognitive Bias. *Transactions of the Association for Computational Linguistics (TACL)*. 2024.
- [11] Dotan, E., G. Jaschek, T. Pupko, and **Y. Belinkov**. Effect of Tokenization on Transformers for Biological Sequences. *Bioinformatics*. 2024.
- [12] Dotan, E., E. Wygoda, N. Ecker, M. Alburquerque, O. Avram, **Y. Belinkov**, and T. Pupko. BetaAlign: a deep learning approach for multiple sequence alignment. *Bioinformatics*. 2025.
- [13] Hanna, M., **Y. Belinkov**, and S. Pezzelle. Are formal and functional linguistic mechanisms dissociated in language models?. *Computational Linguistics*. 2025.

- [14] Mueller, A., J. Brinkmann, M. Li, S. Marks, K. Pal, N. Prakash, C. Rager, A. Sankaranarayanan, A. S. Sharma, J. Sun, E. Todd, D. Bau, and **Y. Belinkov**. The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP Through the Lens of Causal Mediation Analysis. *Computational Linguistics*. 2025.
- [15] Dotan, E., I. Lyubman, E. Bacharach, T. Pupko, and **Y. Belinkov**. BetaDescribe: Providing Rich Descriptions from Protein Sequences. *Proceedings of the National Academy of Sciences*. 2026.

Refereed Conference Papers

- [16] Sajjad, H., K. Darwish, and **Y. Belinkov**. Translating Dialectal Arabic to English. In: *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2013.
- [17] **Belinkov, Y.** and J. Glass. Arabic Diacritization with Recurrent Neural Networks. In: *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2015.
- [18] Romeo, S., G. Da San Martino, A. Barrón-Cedeño, A. Moschitti, **Y. Belinkov**, W. Zhu, Y. Zhang, M. Mohtarami, and J. Glass. Neural Attention for Learning to Rank Questions in Community Question Answering. In: *Proceedings of the 26th International Conference on Computational Linguistics (Coling)*, 2016.
- [19] Adi, Y., E. Kermany, **Y. Belinkov**, O. Lavi, and Y. Goldberg. Fine-grained Analysis of Sentence Embeddings Using Auxiliary Prediction Tasks. In: *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.
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- [128] Grand, G. and **Y. Belinkov**. Adversarial Regularization for Visual Question Answering: Strengths, Shortcomings, and Side Effects. In: *Proceedings of the 2nd Workshop on Shortcomings in Vision and Language (SiVL at NAACL-HLT, Best paper award)*, 2019.
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- [135] Bamberger, Z., O. Glick, Ch. Baskin, and **Y. Belinkov**. DEPTH: Discourse Education through Pre-Training Hierarchically. In: *Proceedings of the 9th Workshop on Representation Learning for NLP (RepL4NLP)*, 2025.
- [136] Nikankin, Y., D. Arad, I. Itzhak, A. Reusch, A. Simhi, G. Kesten-Pomeranz, and **Y. Belinkov**. BlackboxNLP-2025 MIB Shared Task: Improving Circuit Faithfulness via Better Edge Selection. In: *Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP (BlackboxNLP)*, 2025.

Edited Collections

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- [138] Alishai, A., **Y. Belinkov**, G. Chrupała, D. Hupkes, Y. Pinter, and H. Sajjad. Proceedings of the 2020 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP (held in EMNLP 2020).
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- [141] **Y. Belinkov**, S. Hao, J. Jumelet, N. Kim, A. McCarthy, and H. Mohebbi. Proceedings of the Sixth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP (held in EMNLP 2023).
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- [143] **Y. Belinkov**, A. Mueller, N. Kim, H. Mohebbi, H. Chen, D. Arad, and G. Sarti. Proceedings of the 8th BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP (held in EMNLP 2025).

Non-Refereed Conference Papers

- [144] **Belinkov, Y.** Large-Scale Electronic Corpora and the Study of Middle and Mixed Arabic. In: *Proceedings of the IVth AIMA International Conference (Emory University, Atlanta, GA, USA, 12–15 October 2013)*, 2021.
- [145] Arad, D., **Y. Belinkov**, H. Chen, N. Kim, H. Mohebbi, A. Mueller, G. Sarti, and M. Tutek. Findings of the BlackboxNLP 2025 Shared Task: Localizing Circuits and Causal Variables in Language Models. In: *Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP (BlackboxNLP)*, 2025.

16. CONFERENCES

Conference Co-Organizer

The Israeli Seminar on Computational Linguistics (*ISCOL* 2021)

17. SELECTED TALKS

- 2026 AI Interpretability: From Insight to Action – Computer Science Colloquium, Harvard University; *Keynote Talk*, AI Transparency Conference, Nuremberg, Germany
- 2026 How and Why Does Localization Emerge in Deep Neural Nets? – Statistical Physics for Ambitious Interpretability Workshop
- 2025 Scientific Insights from Interpretability Research: Some Opportunities and Challenges – AI & Scientific Discovery Online Seminar by University of Chicago
- 2025 Toward Scalable and Actionable Interpretability – COLM LLM Explainability Workshop, New York University, Princeton University, Cornell Tech, Northeastern University, IVADO Workshop, TTIC
- 2025 Interpreting and Controlling AI: From LLMs to PLMs – Kempner Institute, Harvard University
- 2025 Interpreting and Fixing Text-to-Image Models – Bar Ilan University
- 2025 What do image generators see when they generate images? [In Hebrew] – TECHNOVATION Conference, Technion
- 2024 Recent Progress in Interpreting and Controlling Language Models – Tel Aviv University
- 2024 Interpreting Emergent Communication – Simons Institute
- 2024 Editing Knowledge in Text-to-Image Generation Models – NVIDIA Tel Aviv Lab, Berkeley University
- 2023 On Localization in Language Models – Google Research India, IBM Research Tel Aviv, Google Research Tel Aviv, TU Darmstadt, Düsseldorf University, Bar Ilan University, Simons Institute, Microsoft Research New England
- 2023 Editing Text-to-Image Generation Models – AI Summit, Israel
- 2022 Out-of-Distribution NLP – Hebrew University, Israeli Statistical Association
- 2021 Interpretability and Robustness in Natural Language Processing – AAAI New Faculty Highlights (video)
- 2020 Studying the History of the Arabic Language: Language Technology and a Large-Scale Historical Corpus – The Open University (video)
- 2020 Interpretability and Other Highlights from NLP – Workshop on Decoding Communication in Nonhuman Species, Simons Institute, UC Berkeley
- 2020–21 Causal Mediation Analysis for Interpreting Neural NLP: The Case of Gender Bias – Stanford, UC Berkeley, UMass Amherst, Google, Salesforce, Amazon, NYU, Edinburgh
- 2019 Deep Learning Models for Language: What they learn, where they fail, and how to make them more robust – Hebrew University, Technion, Weizmann Institute, Carnegie Mellon University, University of Pennsylvania
- 2018 Internal Representations in Neural Machine Translation – Amazon MT team, Pittsburgh
- 2018 Internal Representations in Deep Learning for Language and Speech Processing – Johns Hopkins University, University of Washington, Allen Institute for Artificial Intelligence, Toyota Technological Institute at Chicago, Radcliffe Institute for Advanced Study
- 2017 Understanding Internal Representations in Deep Learning Models for Language and Speech Processing – Machine Learning for Language, NYU, New York
- 2017 On Learning Form and Meaning in Neural Machine Translation Models – Computational Data Science Seminar, Technion; CompLang Discussion Group, MIT
- 2017 What do Neural Machine Translation Models Learn about Morphology? – Data Science Summit Europe, Jerusalem
- 2017 Language Technologies for Arabic: Historical Documents, Web Forums, and Machine Translation – Qatar Computing Research Institute, Doha

- 2016 A Computational Analysis of Judeo-Arabic Translations of the Passover Hagaddah – International Jewish Languages Conference, Hebrew University of Jerusalem, Jerusalem
- 2015 Deep Learning for Sentence Representation – IBM Research, Tel Aviv
- 2015 Exploring Compositional Architectures and Word Vector Representations for Prepositional Phrase Attachment – Tel Aviv University, Tel Aviv